

Visible Light-Induced Synthesis of Substituted 6-aryl-8,9-dihydrobenzo[c]phenanthridine-10(7H)-one derivative using disodium salt of eosin Y

A

B. Tech. Project Report Submitted

As Partial Fulfillment of the Requirements for the Degree of
Bachelor of Technology

CH-498

Project II



By

BIJIRGIRI BASUMATARY

Roll No. 200122019

Supervisor: Prof. Subhas Chandra Pan

**Department of Chemistry
Indian Institute of Technology Guwahati
Guwahati-781039, Assam
India**

CERTIFICATE

This is to certify that Mr. Bijirgiri Basumatary having Roll No. 200122019 has worked in my research group as a project entitled “Visible Light-Induced Synthesis of Substituted 6-aryl-8,9-dihydrobenzo[c]phenanthridine-10(7H)-one derivative using disodium salt of Eosin Y” under my supervision at the Department of Chemistry, Indian Institute of Technology Guwahati, Assam. The work reported herein is an outcome of his own investigation carried out during the semester project work in my laboratory.

I hereby confirm the originality of the work and recommend her the submission of this report to the Department of Chemistry, Indian Institute of Technology Guwahati as a partial fulfillment of the requirements for the degree of B. Tech in Chemistry.

Prof. Subhas Chandra Pan
(Project Supervisor)

IIT Guwahati, India
16th April, 2024

DECLARATION

I declare that the work presented in this project titled “*Visible light-induced synthesis of substituted 6-aryl-8,9-dihydrobenzo[c] phenanthridine-10(7H)-one derivative using through disodium salt of eosin Y*,” submitted to the Department of Chemistry, and to **Prof. Subhas Chandra Pan**, Indian Institute of Technology Guwahati, for the project report of BTP (Phase-II), is my original work. I have not plagiarized or submitted the same work for the project report.

Bijirgiri Basumatary
200122019

ACKNOWLEDGEMENT

I express my profound gratitude to my research supervisor **Prof. Subhas Chandra Pan** for his guidance, patience, valuable educative discussion and continuous help in all possible ways. Without his guidance and persistent help, this project would not have been possible.

I also owe a great debt of gratitude to the research scholars of my lab **Mr. Mukesh Kumar** for his constant support, timely suggestions and inspirations with kindness and enthusiasm have enabled me to complete my project.

Bijirgiri Basumatary

200122019

CONTENTS

1. Abstract
2. Introduction
3. Previous Work and Current Work
4. Optimization Reaction Conditions
5. Mechanism
6. Result and Discussion
7. Experimental section
8. Characterisation of the Product
9. Conclusions
10. References

ABSTRACT

Organic reactions induced by visible light, specifically benzylic oxidation, are significant chemical transformations in organic chemistry, and the use of suitable photocatalysts is crucial to their efficiency. However, costly transition-metal complexes and elaborate organic dyes are the most often utilized photocatalysts, which makes large-scale manufacture difficult. Here, we present a novel strategy for regioselective benzylic oxidation for the greener synthesis of 6-aryl-8,9-dihydrobenzo[*c*]phenanthridin-10(7H)-ones from benzo[*c*]-chromeno[4,3,2-*gh*]phenanthridines in the presence of disodium salt of eosin Y as photocatalysts under irradiation of 2 x 9 W white LED light as shown in (Scheme 1d). The technique is incredibly effective, affordable, sustainable and eco-friendly tactic, and the disodium salt of eosin Y with light symbolize a development in photochemistry.

INTRODUCTION

Visible light photocatalysis has drawn a lot of interest in the last few years due to its affordability, ease of use, and environmental friendliness, nonetheless the photocatalysts¹ has significant role on the reaction's efficiency. The most popular photocatalysts are organic dyes and transition metal complexes; however, their relatively high cost makes them less conducive². Our innovative strategy is to examine an *in-situ* generated photocatalyst that facilitates the relevant organic synthesis. The regioselective oxidation of saturated carbon and hydrogen (C-H) bonds of widely accessible alkanes to chemicals using atmospheric oxygen represents a great achievement in the research fields³ as well as in the industrial field. Indeed, in nature enzyme-catalyzed process is frequently observed. The regioselective aerobic oxidation of saturated carbon-hydrogen(C-H) bonds is challenging in chemistry⁴. Ketones are important structural motifs and useful intermediates that are present in medications, natural goods, and sophisticated organic materials. According to current research on photocatalysis with visible light, a novel and practical approach for the regioselective oxidation on saturated carbon and hydrogen (C-H) bonds⁵. Due to adjustable photophysical and redox properties, Polycyclic Heteroaromatic Compounds (PAHs) are interesting materials for potential applications in organicelectronics⁶. These polycyclic aromatic hydrocarbons (PAHs) are in demand for new materials in variety of

scientific and technical domains, including solar cells, organic photovoltaics (OPVs), organic light-emitting diodes (OLEDs), and organic field effect (OFELTs). Therefore, Chemists are still trying to synthesize new molecules with better electronic properties.

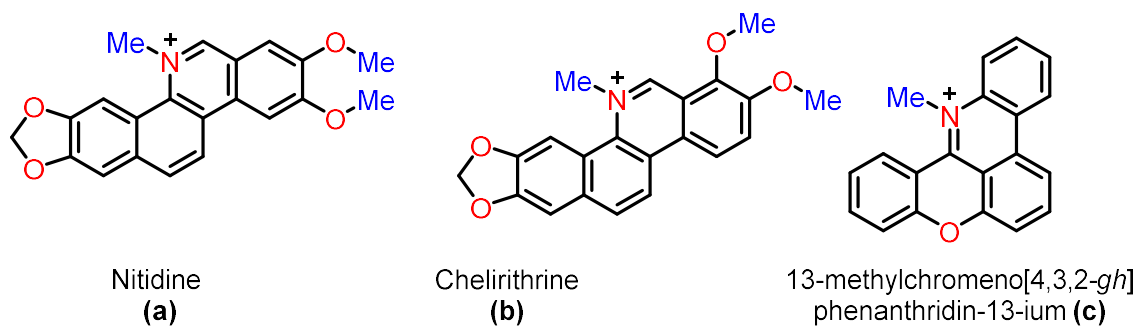


Figure 1 Some biologically important polycyclic heteroaromatic compounds (a-c).

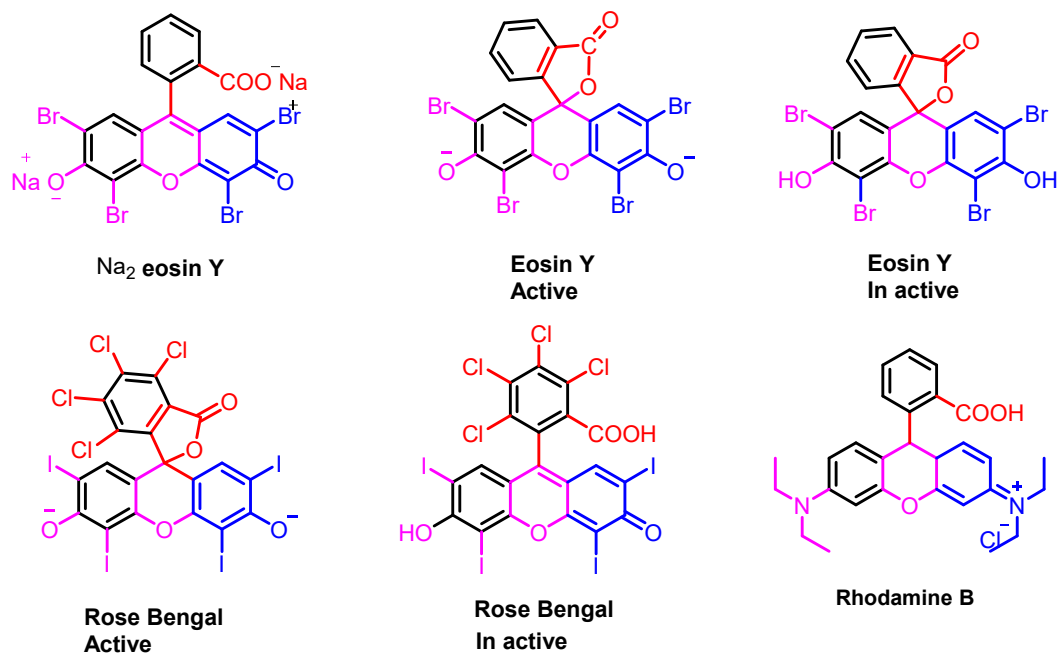
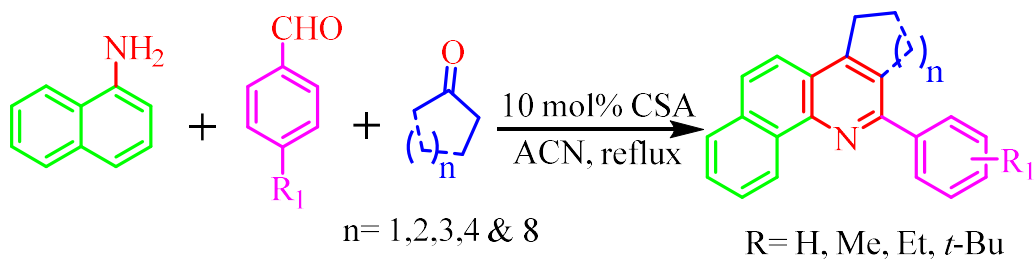


Figure 2 Examples of well-known photochemicals.

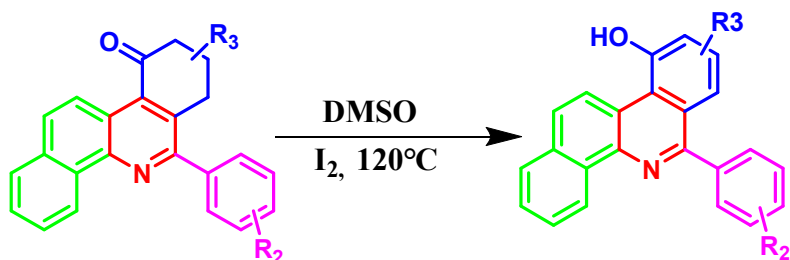
Our group recently demonstrated an approach of synthesis of phenanthridines derivative, as shown below.

Previous Work



J. Org. Chem. 2017, **82**, 12416-12429

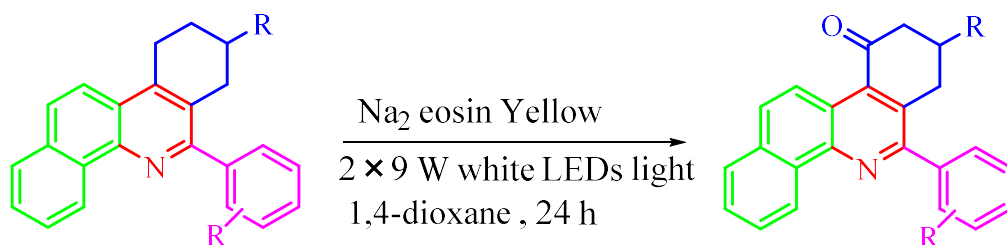
Scheme 1a



Org. Biomol. Chem. 2022, **20**, 7302-7315.

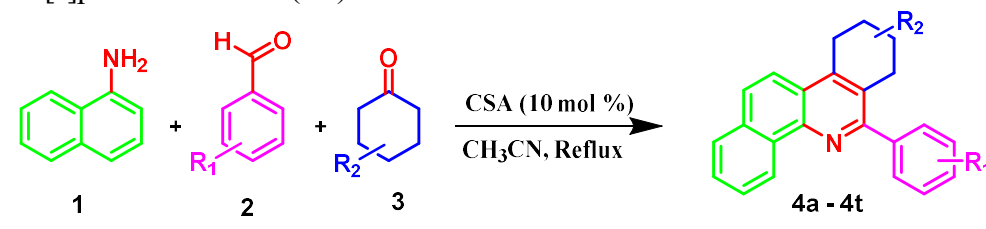
Scheme 1b

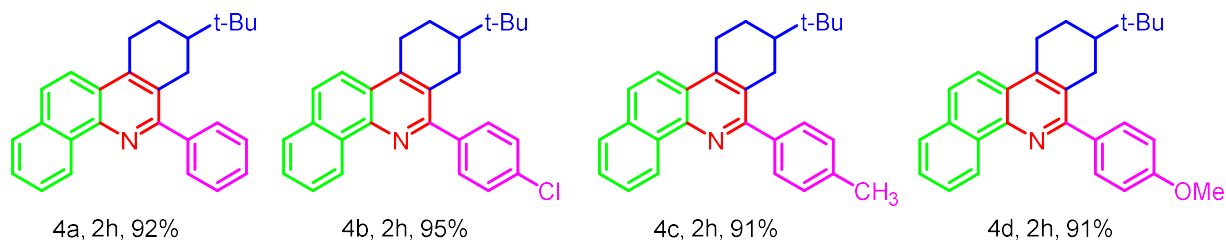
Current Work



Scheme 1c

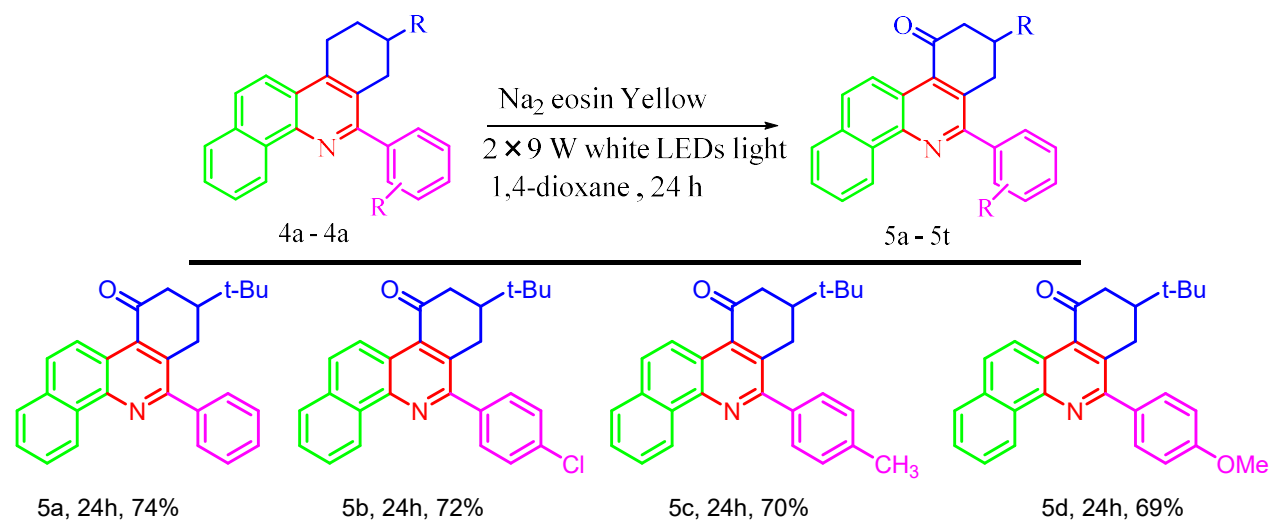
Scheme 1 (1a to b) Previous works; and **(1c)** present work for the synthesis of 6-aryl-8,9-dihydrobenzo[c]phenanthridin-10 (7H)-one derivative.





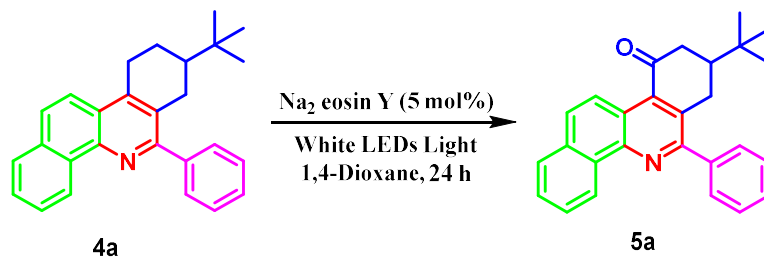
^aAll the reactions were carried out with 1-naphthylamine (**1**, 1 mmol), derivatives of benzaldehyde (**2**, 1 mmol), and cyclohexanone (**3**, 1 mmol) in 2.5 Ml acetonitrile solvent at 81°C for 2 h. ^{an} isolated yield.

Synthesis of 6-aryl-8,9-dihydrobenzo[c]phenanthridin-10(7H)-one derivative



^aAll reactions were carried out by using benzo[c]-chromeno[4,3,2-*gh*]phenanthridines (0.2 mmol), and Na₂ eosin Yellow (5 mol%) in 1.5 Ml 1,4-dioxane and stirred under the irradiation of 2 × 9 W white LEDs bulbs for 24 h. ^b isolated yield.

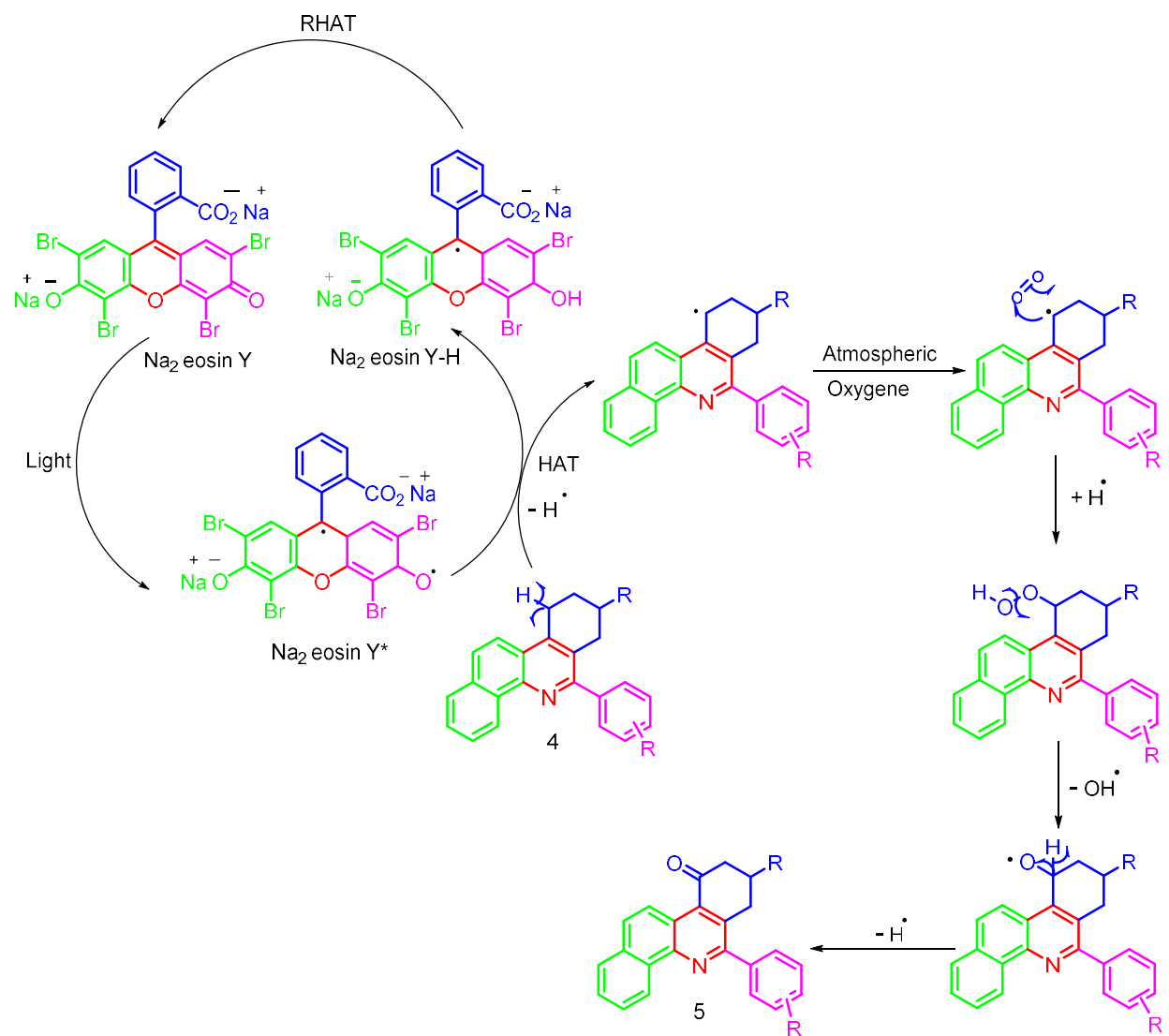
OPTIMIZATION REACTION CONDITIONS^{a,b}



Entry	Photocatalyst	Visible Light Source	Solvent	Time (Hours)	Yield (%)
1	Na ₂ eosin Y	White LEDs Light (2 × 9 W)	H ₂ O	24	ND
2	Na ₂ eosin Y	White LEDs Light (2 × 9 W)	Toluene	24	ND
3	Na ₂ eosin Y	White LEDs Light (2 × 9 W)	CHCl ₃	24	24
4	Na ₂ eosin Y	White LEDs Light (2 × 9 W)	CH ₃ CN	24	31
5	Na ₂ eosin Y	White LEDs Light (2 × 9 W)	EtOH	24	39
6	Na ₂ eosin Y	White LEDs Light (2 × 9 W)	THF	24	42
7	Na₂ eosin Y	White LEDs Light (2 × 9 W)	1,4-dioxane	24	74
8	Na ₂ eosin Y	White LEDs Light (2 × 9 W)	1,4-dioxane	18	59
9	Na ₂ eosin Y	White LEDs Light (2 × 9 W)	1,4-dioxane	27	74
10	Na ₂ eosin Y	White LEDs Light (1 × 9 W)	1,4-dioxane	24	60
11	Na ₂ eosin Y	White LEDs Light (3 × 9 W)	1,4-dioxane	20	74
12	Na ₂ eosin Y	White LEDs Light (4 × 9 W)	1,4-dioxane	18	74
13	Na ₂ eosin Y	Green LEDs Light (2 × 9 W)	1,4-dioxane	24	70
14	Na ₂ eosin Y	Blue LEDs Light (2 × 9 W)	1,4-dioxane	24	66
15	Na ₂ eosin Y	Dark	1,4-dioxane	24	ND
16	Na ₂ eosin Y	Sun Light (Top of the roof)	1,4-dioxane	12	Trace

^aAll reactions were carried out with benzo[*c*]-chromeno[4,3,2-*gh*]phenanthridines(0.2 mmol) (**4f**), and Na₂ eosin Yellow (5 mol%) in 1.5 mL 1,4-dioxane solvent and stirred under the irradiation of 2 × 9 W white LEDs bulbs for 24 h. ^bIsolated yield.

Mechanism



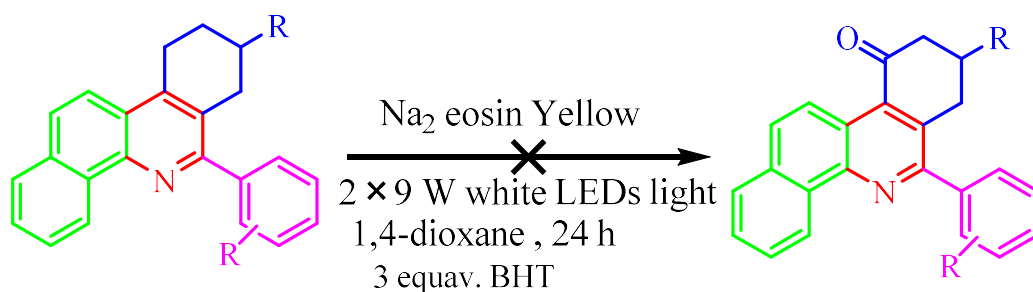
RESULT AND DISCUSSION

We embarked on a meaningful study by initially establishing the optimal experimental conditions. This entailed conducting a model reaction involving substituted benzo[c]-chromeno[4,3,2-gh]phenanthridines under visible light irradiation in the presence of organic dye such as Na₂ eosin yellow photocatalyst with 1,4-dioxane solvent for 24 h and maintain the temperature 25 to 30 C as indicated in the optimization table. The solvent and the reaction time were screened, and the yield was determined using ¹H NMR with area normalization. From entries 1 to 2, the reactions did not happen in which the solvents were H₂O and toluene. From entries no. 3 to 4, reactions happened in which the solvents were CHCl₃ and CH₃CN, and the yield was 24% and 31%, respectively. Again, the entry no. 5 to 6, the reactions happened in which the solvents were EtOH and THF, the yield was 39%, 42%, and similarly, from entry no. 7 to 9, when I was changed in a variation of times 24, 18, and 27 h, the change in percentage yield of the products was 74%, 59%, and 74%. In entries no. 10 to 12, when I changed visible light power the change in the percentage yield of the product was from 60% to 74%. In entry no. 13 to 14, when I was changing from green to blue the percentage yield of the product was from 70% to 66%. From entries no. 15 to 16 in dark conditions, the reactions did not happen, but in sunlight, the change of yield of the reaction was 22%. Na₂ eosin yellow was the most favorable photocatalyst, the suitable solvent was 1,4-dioxane, and 2×9 W white LED bulbs delivered the desired product in 74% yield. This result indicates that 1,4-dioxane solvent proved the most effective (entries 1-16). Interestingly, the yield of 5a was significantly increased to 74% when the reaction was carried out in an oven-dried 10 mL screw-cap borosilicate vial by using Na₂ eosin yellow (5 mol%) and a solvent such as 1,4-dioxane 1.5 ml. Subsequently, the model reaction was tested under 2×9 W white, blue, and green LED bulbs in oven-dried 10 mL screw-capped borosilicate vial, affording. Further investigation regarding the quantity of photocatalyst and solvent was finally conducted (for details, see Table 1). Following intensive experimentation, the optimal conditions were determined as follows substituted benzo[c]-chromeno[4,3,2-gh]phenanthridines 1 (0.2 mmol %), Na₂ eosin Y (5 mol%), 1,4-dioxane solvent (1.5 mL) under the irradiation of 2×9 White LEDs bulbs for 24 h, With the optimized conditions at our disposal, we proceeded to investigate further the potential applications of this innovative the visible-light-induced method.

EXPERIMENTAL SECTION

To an oven-dried 10 mL screw-cap borosilicate vial, Three-component product benzo[c]-chromeno[4,3,2-gh]phenanthridines (1) (0.2 mmol), Na₂ eosin yellow (2) (5 mol %) were dissolved in 1,4-dioxane (1.5 mL), and then the reaction mixture was stirred under the irradiation of 2×9 W white LEDs bulbs (400 to 700 nm) at a distance of approximately 2-4 cm for 24 h. After completion of the reaction (monitored by TLC analysis), the reaction mixture was diluted with brine solution (5 mL) and extracted with ethyl acetate (12 mL). The combined organic layer was dried over anhydrous Na₂SO₄, and the solvent was evaporated under reduced pressure. The crude residue thus obtained was purified by column chromatography over silica gel (60-120 mesh) using only hexane as eluent to afford the desired products-6-aryl-8,9-dihydrobenzo[c]phenanthridin-10 (7H)-ones in 74 yield. Each set of a reaction (entry 1-18) was also performed similarly with almost identical yields. Each of the synthesized compounds was structurally confirmed by spectroscopy studies, including ¹H NMR, ¹³C NMR, IR, Melting points, and HRMS spectroscopic technique.

Control Experiment



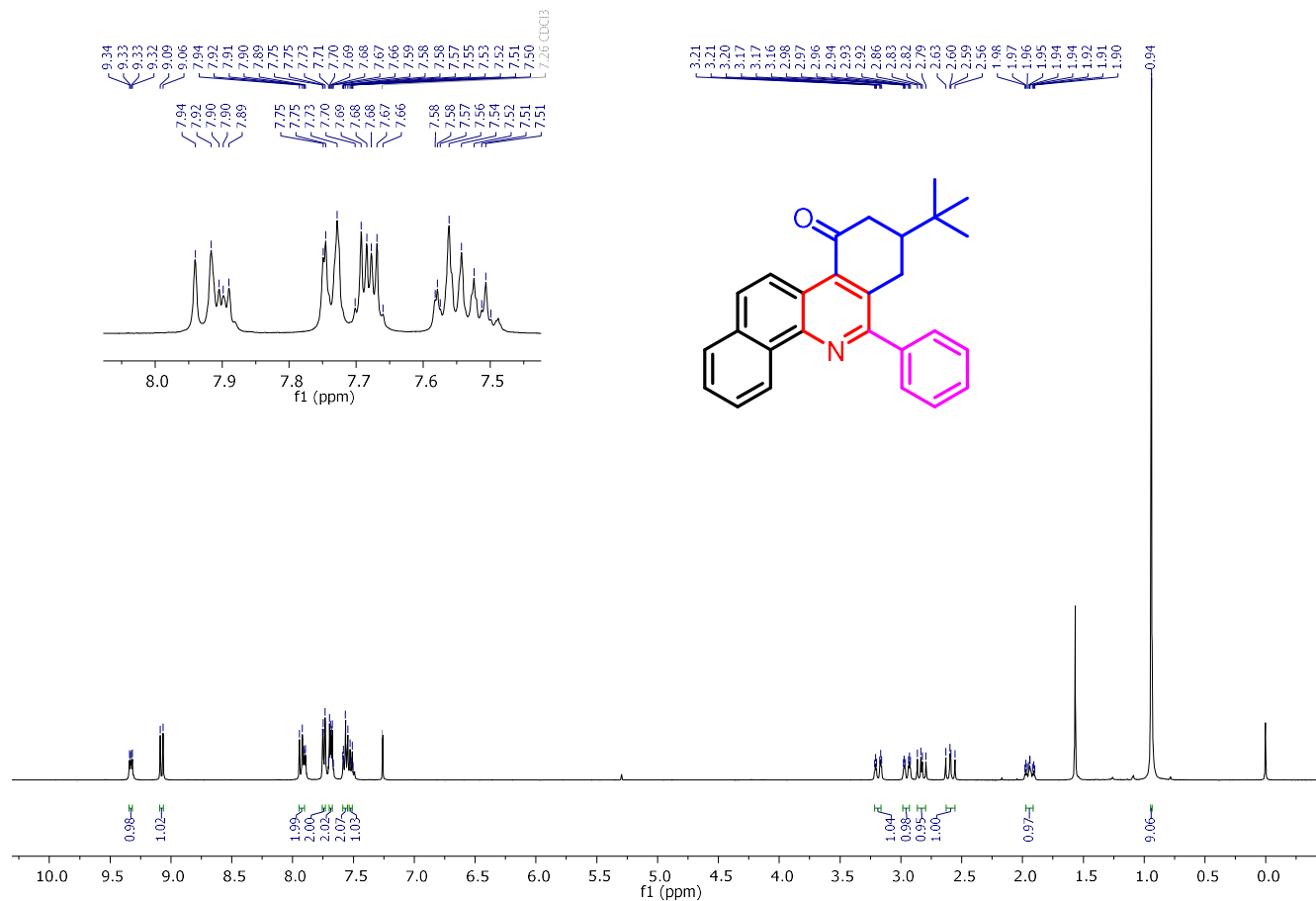
CONCLUSIONS

As a tool for organic synthesis, visible light photo redox catalyst with metal free has already received a lot of attention. Disodium salt of eosin Y functions as an attractive alternative for redox active metal free for a variety of applications, sometimes even outperforming them. And it has been applied to generate reactive intermediates such as aryl radical, iminium ion, enone radical anion, hetero cyclization, aromatization amine functionalization hydroxylation, reduction and oxidation reaction, etc. The photocatalyst Na₂ eosin Y has got much more attention due to its easy handling, less expensive, eco-friendly, non-toxic, etc. In addition, Na₂ eosin Y catalysis has been merged with other modes of catalyst, such as enamine catalyst and hydrogen bond promoted catalyst, to achieve an enantioselective reaction. The use of Na₂ eosin Y photocatalysis in continuous new technology has been described. Overall, the good availability, strong absorption in the visible part of the spectrum, and suitable redox potential value for various organic transformations make Na₂ eosin Y an appealing and green photocatalyst for organic synthesis. A series of control experiments were carried out to expound the mechanism of the process (Scheme 1b). I saw no reaction in product **5a** in the presence of 2 equiv. of radical scavenger (BHT or TEMPO). And the raw material 4a was nearly quantitatively recovered (Scheme 1c). Control experiments under an oxygen atmosphere were performed to further investigate whether oxygen is involved in this process (scheme 1c).

SPECTRAL DATA

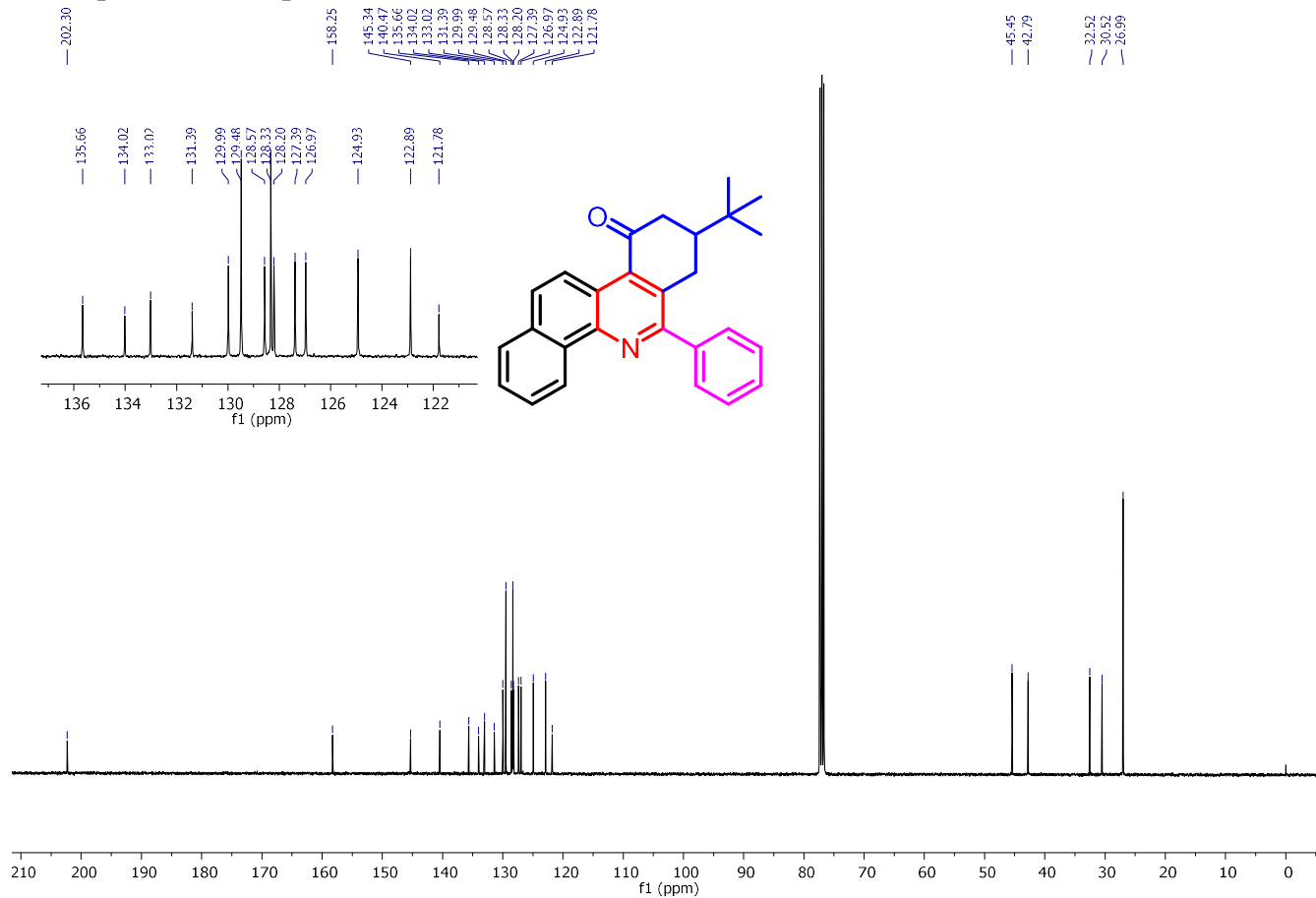
¹H NMR of 5a

ATK-MK-G-1-1H.1.fid — ATK-MK-G-1-1H



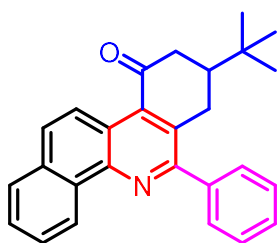
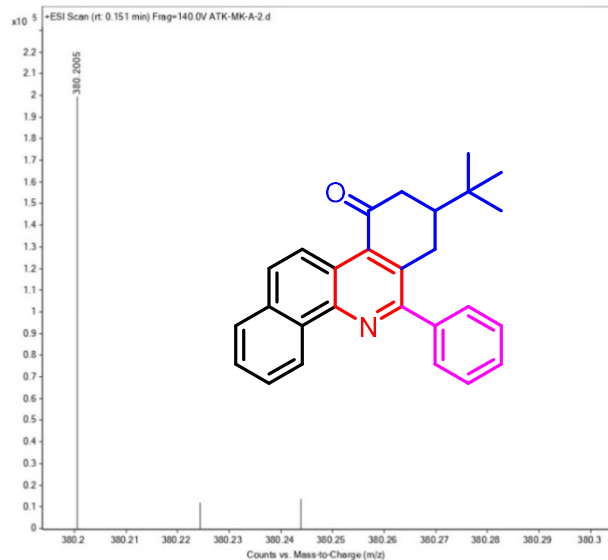
¹³C NMR of 5a

ATK-MK-2-A_13C.12.fid — ATK-MK-2-A_13C



HRMS of 5a

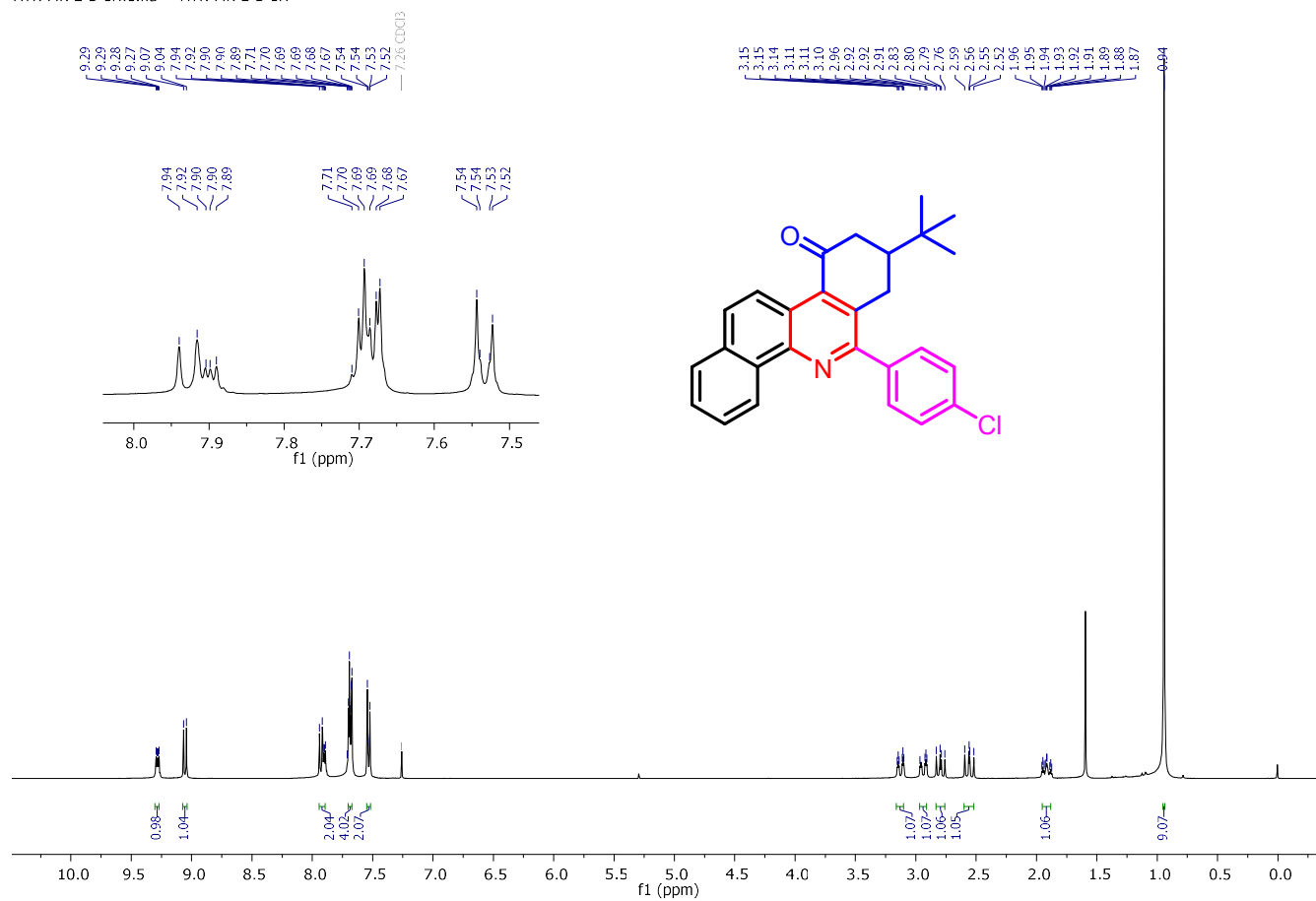
Sample Name	Sample17	Position	P1-B4	Instrument Name	QTOF
User Name	SYSTEM (SYSTEM)	Inj Vol	5	InjPosition	
Sample Type	Sample	IRM Calibration Status	Success	Data Filename	ATK-MK-A-2.d
ACQ Method	DIRECT MASS_POSITIVE_100_1500.m	Comment		Acquired Time	11-09-2023 15:42:41 (UTC+05:30)



8-(tert-butyl)-6-phenyl-8,9-dihydrobenzo[c]17henanthridine-10(7H)-one (5a). Yield 74% (56.5 mg), solid, white colour, mp 213-215°C; ^1H NMR (400 MHz, Chloroform-*d*) δ 9.32 (dd, J = 6.0, 3.6 Hz, 1H), 9.07 (d, J = 9.4 Hz, 1H), 7.94 – 7.90 (m, 2H), 7.75 – 7.73 (m, 2H), 7.68 (dd, J = 6.1, 3.3 Hz, 2H), 7.58 – 7.54 (m, 2H), 7.53 – 7.50 (m, 1H), 3.21 – 3.16 (m, 1H), 2.98 – 2.93 (m, 1H), 2.83 (d, J = 4.4 Hz, 1H), 2.62 – 2.55 (m, 1H), 1.94 (td, J = 10.1, 9.4, 6.1 Hz, 1H), 0.94 (s, 9H); ^{13}C NMR (101 MHz, Chloroform-*d*) δ 158.25, 140.47, 135.66, 134.02, 133.02, 131.39, 129.99, 128.57, 128.33, 128.20, 127.39, 126.97, 124.93, 122.89, 45.45, 42.79, 32.52, 30.52, 26.99. IR(KBr) ν_{max} 1729, 2979; HRMS (ESI) calcd for $\text{C}_{27}\text{H}_{25}\text{NO}$ 380.2009 ($\text{M}+\text{H}^+$); Found 380.2006

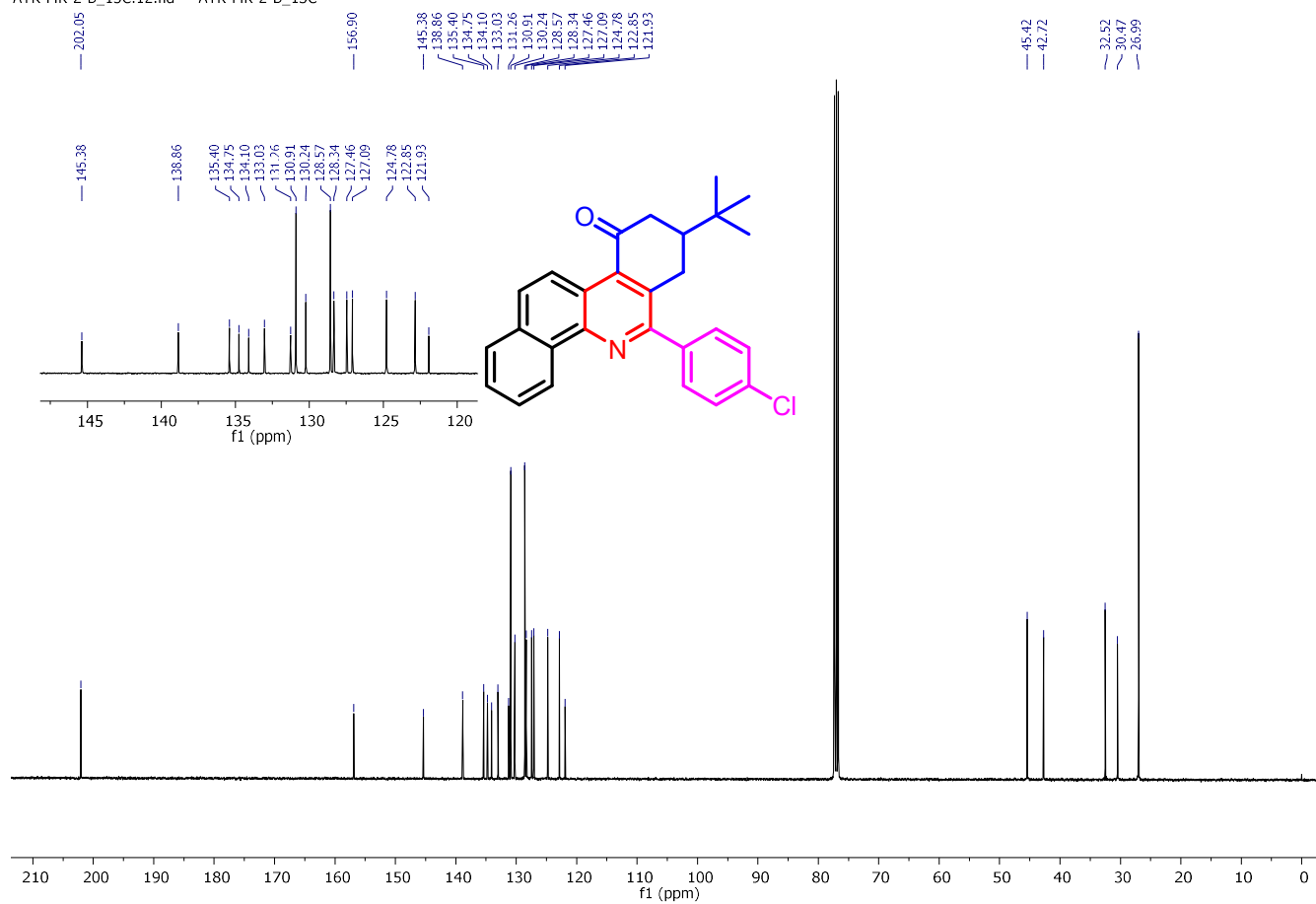
¹H NMR of 5b

ATK-MK-2-B-1H.1.fid — ATK-MK-2-B-1H



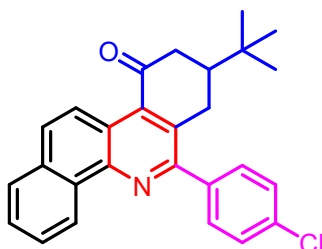
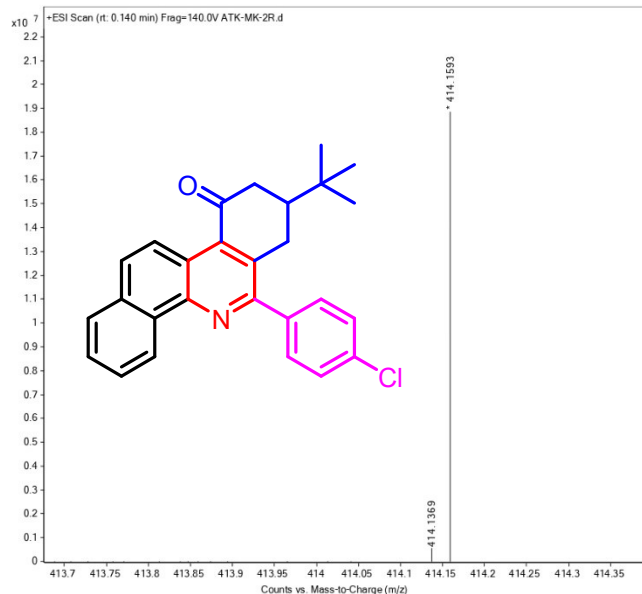
¹³C NMR of 5b

ATK-MK-2-B_13C.12.fid — ATK-MK-2-B_13C



HRMS of 5b

Sample Name	Sample10	Position	P1-A9	Instrument Name	QTOF
User Name	SYSTEM (SYSTEM)	Inj Vol	5	InjPosition	
Sample Type	Sample	IRM Calibration Status	Success	Data Filename	ATK-MK-2R.d
ACQ Method	DIRECT MASS_POSITIVE_100_1500.m	Comment		Acquired Time	16-10-2023 10:03:33 (UTC+05:30)

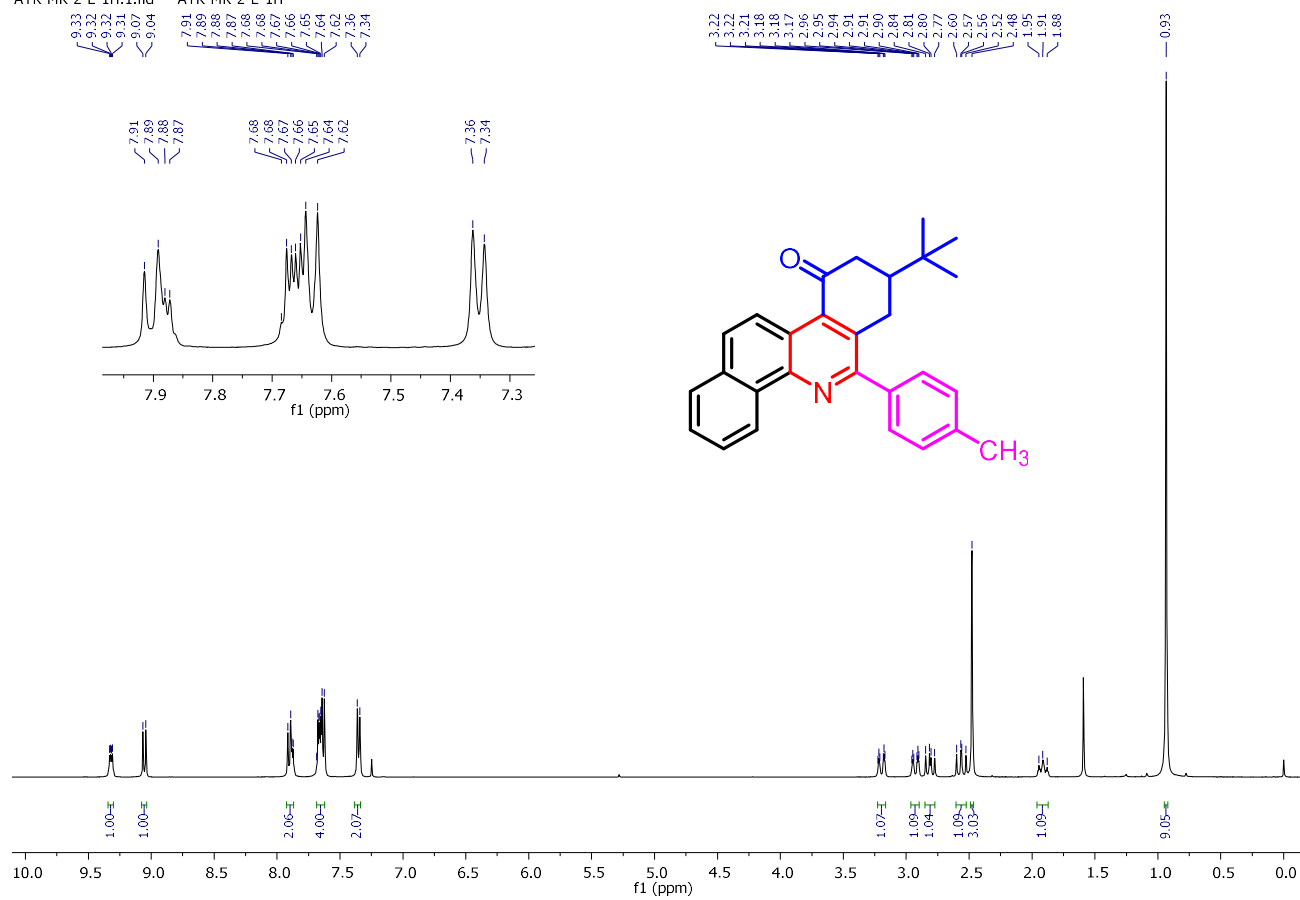


8-(tert-butyl)-6-(4-chlorophenyl)-8,9-dihydrobenzo[c]phenanthridin-10(7H)-one (5b).

Yield 70% (60 mg). solid, white colour, mp 240°C: ^1H NMR (400 MHz, Chloroform-*d*) δ 9.28 (dd, $J = 6.0, 3.5$ Hz, 1H), 9.05 (d, $J = 9.4$ Hz, 1H), 7.93 (d, $J = 9.6$ Hz, 2H), 7.70 – 7.67 (m, 4H), 7.55 – 7.52 (m, 2H), 3.16 – 3.10 (m, 1H), 2.97 – 2.91 (m, 1H), 2.83 – 2.76 (m, 1H), 2.60 – 2.52 (m, 1H), 1.95 – 1.88 (m, 1H), 0.94 (s, 9H).; ^{13}C NMR (101 MHz, Chloroform-*d*) δ 202.05, 138.86, 135.40, 134.75, 134.10, 133.03, 131.26, 130.91, 130.24, 128.57, 128.34, 127.46, 127.09, 124.78, 122.85, 121.93, 45.42, 42.72, 32.52, 30.47, 26.99; IR(KBr) ν_{max} 1728, 2940; HRMS (ESI) calcd for $\text{C}_{27}\text{H}_{24}\text{ClNO}$ 414.1619 ($\text{M}+\text{H}^+$); Found 414.1593

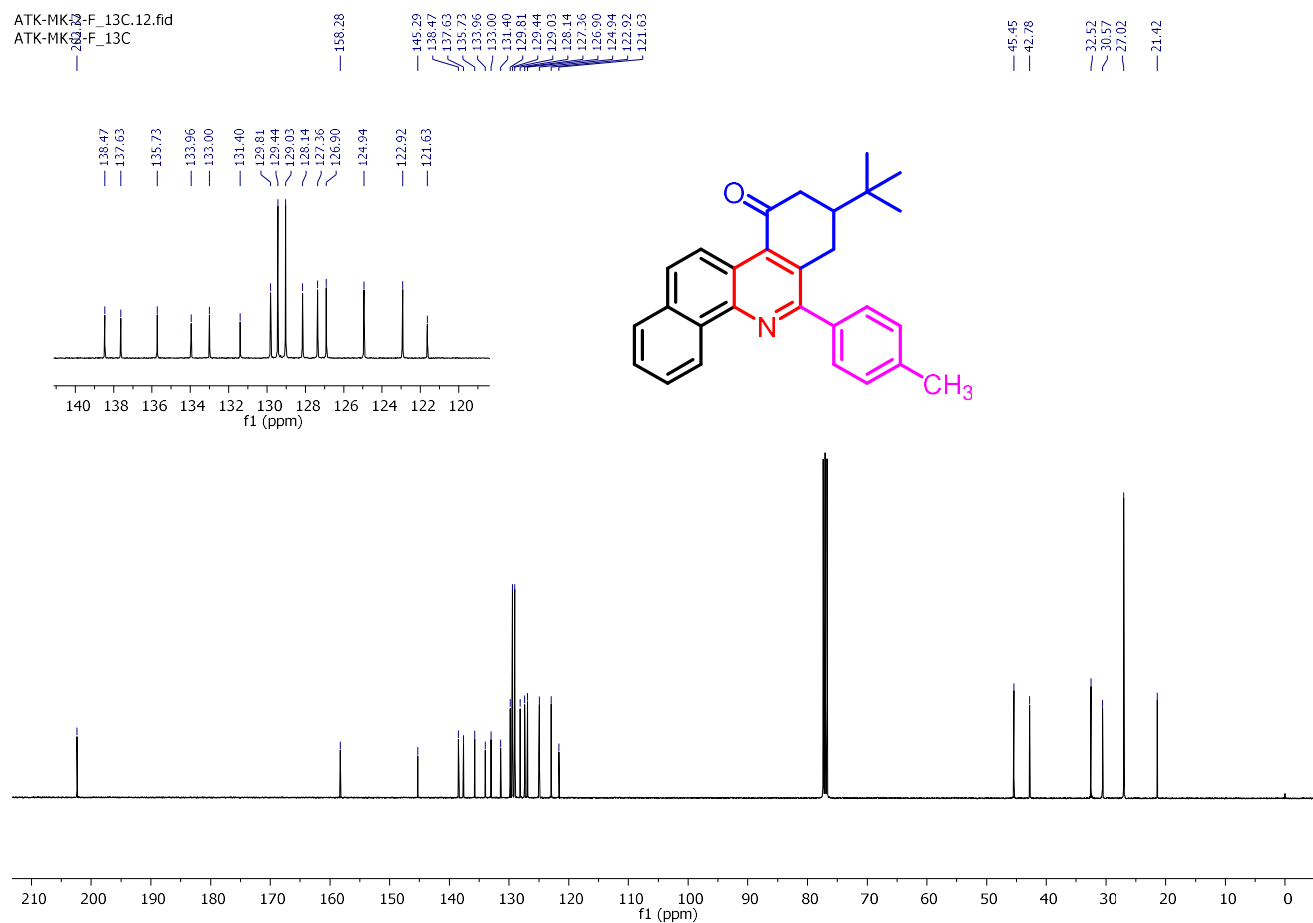
¹H NMR of 5c

ATK-MK-2-E-1H.1.fid — ATK-MK-2-E-1H



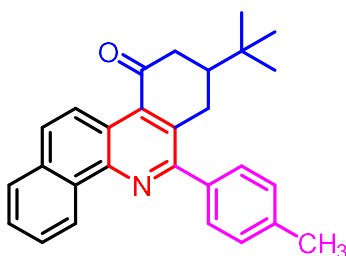
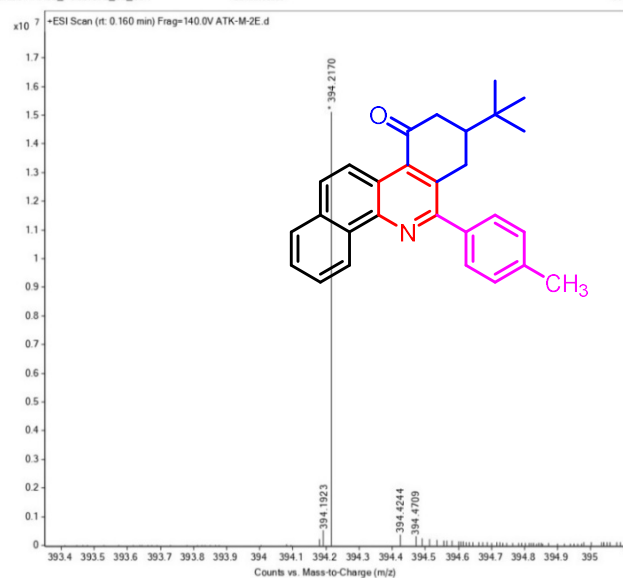
¹³C NMR of 5c

ATK-MK2-F_13C.12.fid
ATK-MK2-F_13C



HRMS of 5c

Sample Name		Position	P1-B6	Instrument Name	QTOF
User Name	SYSTEM (SYSTEM)	Inj Vol	5	InjPosition	
Sample Type	Sample	IRM Calibration Status	Success	Data Filename	ATK-M-2E.d
ACQ Method	DIRECT MASS_POSITIVE_01_1.m	Comment		Acquired Time	31-10-2023 15:17:54 (UTC+05:30)

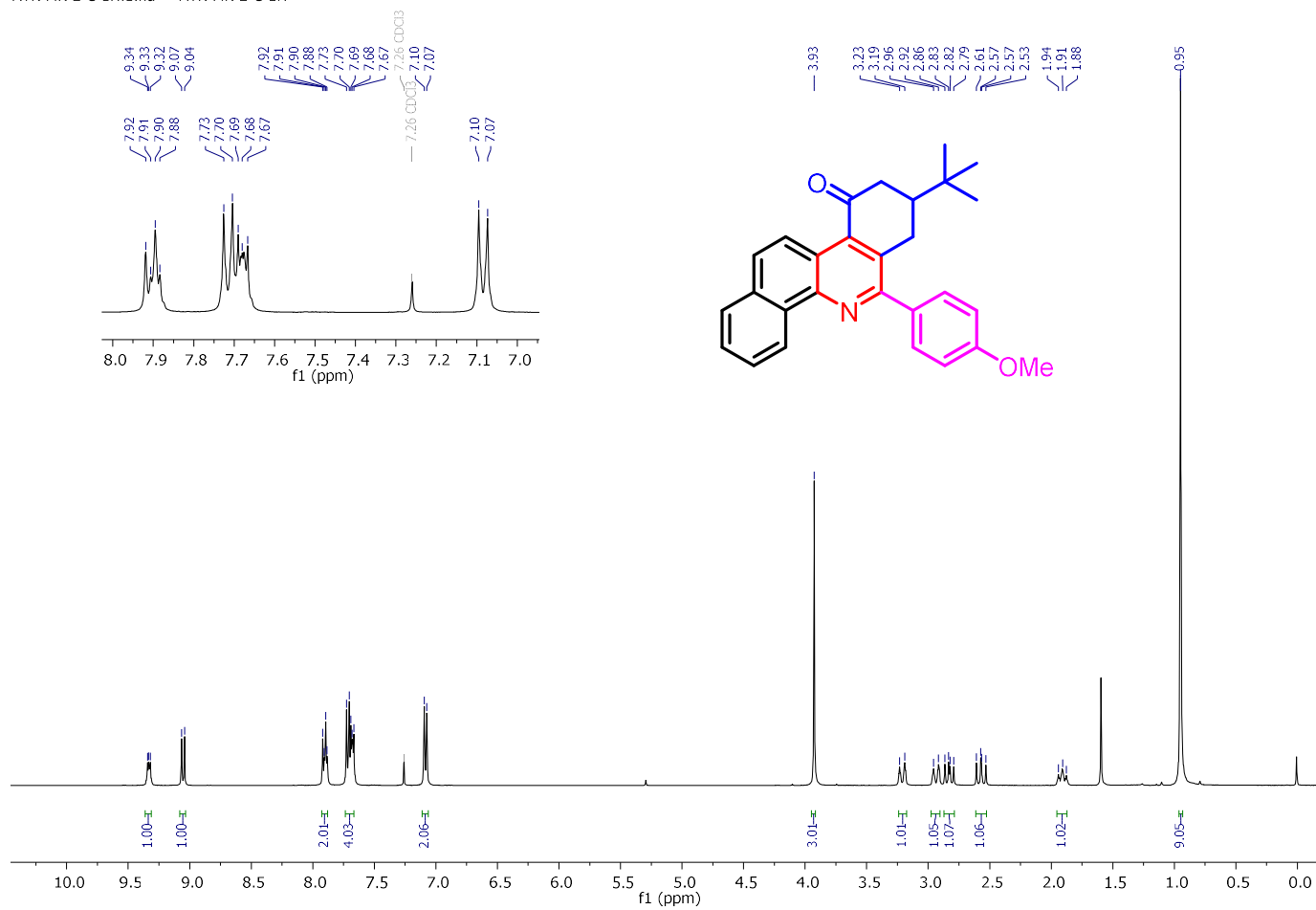


8-(tert-butyl)-6-(p-tolyl)-8,9-dihydrobenzo[c]phenanthridin-10(7H)-one(5c).

Yield 67%(55.5mg). solid, white colour, mp 227-229°C: ^1H NMR (400 MHz, Chloroform-*d*) δ 9.32 (dd, $J = 6.0, 3.5$ Hz, 1H), 9.05 (d, $J = 9.3$ Hz, 1H), 7.93 – 7.87 (m, 2H), 7.69 – 7.62 (m, 4H), 7.35 (d, $J = 7.8$ Hz, 2H), 3.20 (dt, $J = 16.4, 2.7$ Hz, 1H), 2.93 (dt, $J = 16.5, 3.0$ Hz, 1H), 2.81 (dd, $J = 16.4, 12.1$ Hz, 1H), 2.56 (dd, $J = 16.4, 14.0$ Hz, 1H), 2.48 (s, 3H), 1.91 (t, $J = 12.9$ Hz, 1H), 0.93 (s, 9H).; ^{13}C NMR (101 MHz, Chloroform-*d*) δ 202.37, 138.47, 137.63, 135.73, 133.00, 131.40, 129.81, 129.44, 129.03, 127.36, 126.90, 124.94, 122.92, 121.63, 45.45, 42.78, 32.52, 30.57, 27.02, 21.42.; IR(KBr) ν_{max} 1719, 2915: HRMS (ESI)calcd for $\text{C}_{28}\text{H}_{27}\text{NO}$ 394.2165 ($\text{M}+\text{H}^+$); Found 394.2170

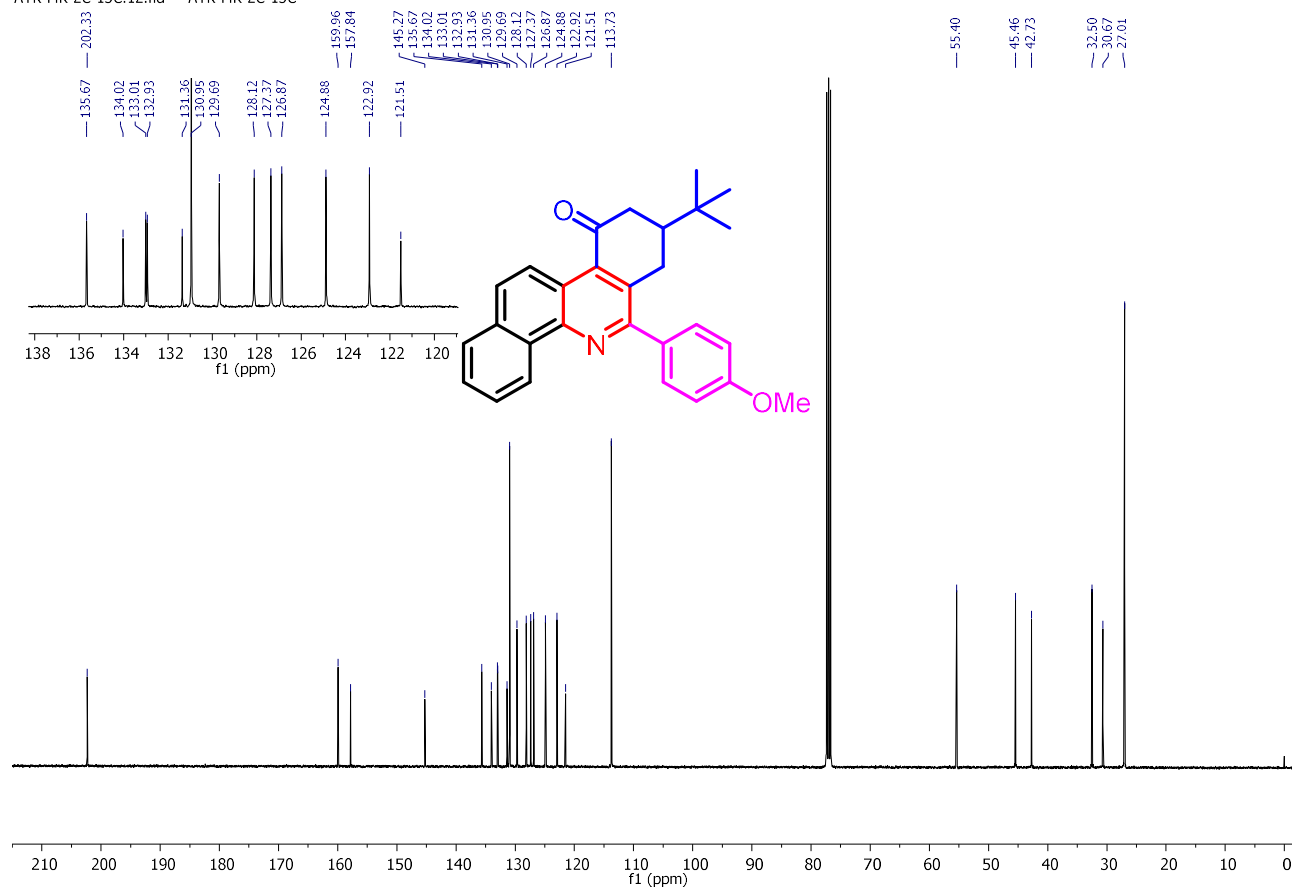
¹H NMR of 5d

ATK-MK-2-C-1H.1.fid — ATK-MK-2-C-1H



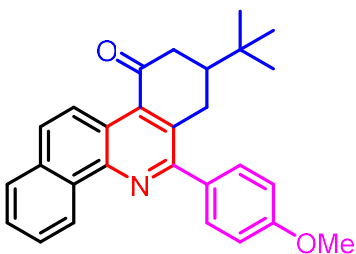
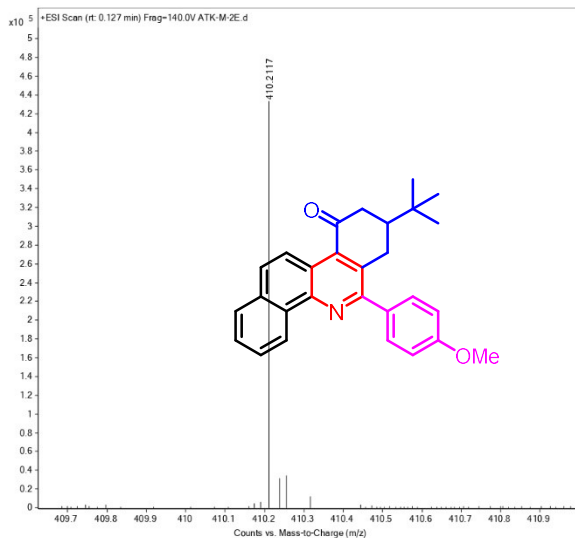
¹³C NMR of 5d

ATK-MK-2C-13C.12.fid — ATK-MK-2C-13C



HRMS of 5d

Sample Name	SYSTEM (SYSTEM)	Position	P1-B6	Instrument Name	QTOF
User Name	Sample	Inj Vol	5	InjPosition	
Sample Type	DIRECT MASS_POSITIVE_01_1.m	IRM Calibration Status	Success	Data Filename	ATK-M-2E.d
ACQ Method		Comment		Acquired Time	31-10-2023 15:17:54 (UTC+05:30)



8-(tert-butyl)-6-(4-methoxyphenyl)-8,9-dihydrobenzo[c]phenanthridin-10(7H)-one (5d).

Yield 69% (57 mg). solid, white colour, mp 234-236°C: ^1H NMR (400 MHz, Chloroform-*d*) δ 9.36 – 9.31 (m, 1H), 9.05 (d, J = 9.4 Hz, 1H), 7.90 (dd, J = 9.2, 4.8 Hz, 2H), 7.74 – 7.67 (m, 4H), 7.08 (d, J = 8.7 Hz, 2H), 3.93 (s, 3H), 3.21 (d, J = 16.3 Hz, 1H), 2.94 (d, J = 16.5 Hz, 1H), 2.83 (dd, J = 16.4, 12.1 Hz, 1H), 2.57 (dd, J = 16.5, 13.9 Hz, 1H), 1.91 (t, J = 12.9 Hz, 1H), 0.95 (s, 9H); ^{13}C NMR (101 MHz, Chloroform-*d*) δ 202.33, 159.96, 157.84, 135.67, 133.01, 132.93, 131.36, 130.95, 129.69, 128.12, 127.37, 126.87, 124.88, 122.92, 113.73, 55.40, 45.46, 42.73, 32.50, 30.67, 27.01; IR(KBr) ν_{max} 1710, 2918; HRMS (ESI) calcd for $\text{C}_{28}\text{H}_{27}\text{NO}_2$ 410.2115 ($\text{M}+\text{H}^+$); Found 410.2117

References

1. M. Tanioka, A. Kuromiya, R. Ueda, T. Obata, A. Muranaka, M. Uchiyama and S. Kamino, *Chem. Commun.*, 2022, 58, 7825-7828.
2. F. Mohamadpour, *ACS Omega*, 2022, 7, 8429-8436.
3. D. P. Hari and B. König, *Chem. Commun.*, 2014, 50, 6688-6699.
4. A. Bosveli, T. Montagnon, D. Kalaitzakis and G. Vassilikogiannakis, *Org. Biomol. Chem.*, 2021, 19, 3303-3317.
5. Y. Song, H. Zhang, J. Guo, Y. Shao, Y. Ding, L. Zhu and X. Yao, *Eur. J. Org. Chem.*, 2021, 2021, 5914-5921.
6. F. Mohamadpour, *Front. Chem.*, 2022, 10, 880257
7. R. Li, X. Chen, S. Wei, K. Sun, L. Fan, Y. Liu, L. Qu, Y. Zhao and B. Yu, *Adv. Synth. Catal.*, 2018, 360, 4807-4813.
8. G. Shukla, A. Dahiya, T. Alam and B. K. Patel, *Asian J. Org. Chem.*, 2019, 8, 2243-2248.
9. A. K. Sahoo, A. Rakshit, A. Dahiya, A. Pan and B. K. Patel, *Org. Lett.*, 2022, 24, 1918-1923.
10. Y. Liu, P. Zhou, Y. Xu, Z. Yang and D. Wang, *Chem. Commun.*, 2023, 59, 1681-1684.
11. M. Jarrahi, B. Maleki and R. Tayebjee, *RSC Adv.*, 2022, 12, 28886–28901.
12. S. Yashmin, R. Ali, S. Mondal and A. T. Khan, *Chem. Commun.*, 2022, 58, 5853-5856.
13. S. Yashmin, S. Mondal, R. Das, P. Banerjee and A. T. Khan, *Org. Biomol. Chem.*, 2022, 20, 7302-7315.
14. R. Gattu, P. R. Bagdi, R. S. Basha and A. T. Khan, *J. Org. Chem.*, 2017, 82, 12416-12429.
15. Yashmin, A. T. Khan, S. J. Akula, R. P and K. Bhattacharyya, *J. Org. Chem.*, 2023, 88, 13622-13633.
16. A. Borissov, Y. K. Maurya, L. Moshniaha, W.-S. Wong, M. Żyła-Karwowska and M. Stępień, *Chem. Rev.*, 2022, 122, 565-788.
17. A. Ambrosi, C. K. Chua, A. Bonanni and M. Pumera, *Chem. Rev.*, 2014, 114, 7150-7188.
18. C. G. Neochoritis, T. Zarganes-Tzitzikas, K. Katsampoxaki-Hodgetts and A.